

1. How to define a distance measure between two values of a nominal or discrete attribute?
2. Why is it important to understand the nature of the missing values?
3. What are the different ways of handling missing values? Give examples of the situations when they are useful.
4. What are the advantages and disadvantages of imputations while handling missing values?
5. When are regression-based and kNN-based imputation useful when handling missing values?
6. What are standard techniques for removing noise in the data?
7. What are the advantages of data normalization? Does it help in handling outliers?
8. What is one-hot encoding and when are they useful?
9. Why is domain knowledge considered a crucial component of data science?
10. Explain the difference between correlation and causation. Can a scatter plot prove causation?
11. What is the difference between one-hot encoding and embedding?
12. Define a random variable, whose expectation is unbounded and prove it.
13. Suppose that we roll twice a fair k -sided die with the numbers 1 through k on the die's faces, obtaining values X_1 and X_2 .
 - a. What is $E[\max(X_1, X_2)]$ and $E[\min(X_1, X_2)]$?
 - b. Show that $E[\max(X_1, X_2)] + E[\min(X_1, X_2)] = E[X_1] + E[X_2]$.
14. Does the linearity of expectation, $E[X+Y]=E[X]+E[Y]$, require the random variables X and Y to be independent? Explain.
15. **[New]** The expected number of inches of rain on Saturday is 4, and on Sunday is 2. There is a 50% chance of rain on Saturday. If it rains on Saturday, there is a 75% chance of rain on Sunday. If it does not rain on Saturday, there is only a 25% chance of rain on Sunday. What is the expected total rainfall (in inches) over the weekend?
16. Explain the difference between $P(A \cap B)$ and $P(A|B)$. Why is the denominator $P(B)$ necessary in the formula for conditional probability?
17. The following approach is often called reservoir sampling. Suppose we have a sequence of items passing by one at a time. We want to maintain a sample of one item with the property that it is uniformly distributed over all the items that we have seen at each step. Moreover, we want to accomplish this without knowing the total number of items in advance or storing all of the items that we see.
 - a. Consider the following algorithm, which stores just one item in memory at all times. When the first item appears, it is stored in the memory. When the k th item appears, it replaces the item in memory with probability $1/k$. Explain why this algorithm solves the problem.

18. We roll a standard fair die over and over. What is the expected number of rolls until the first pair of consecutive sixes appears?
19. Give an example of three random events X , Y , Z for which any pair are independent but all three are not mutually independent.
20. Prove that $E[X^k] \geq (E[X])^k$ for any even integer $k > 1$.
21. When is a z-test for a population mean appropriate to use? What are its main assumptions or limitations?
22. What are the types of errors in hypothesis testing? Explain them with examples.
23. When data is unavailable, what is your conclusion?
24. A professor of the 'Advanced Algorithms' course believes his students have a good grasp of 'Graduate Algorithms', i.e., the average score is more than 70%. He randomly chose 10 students who had scored 62, 80, 75, 95, 59, 70, 63, 68, 90, and 84 in graduate algorithms out of 100. With a 5% level of significance, does the data have enough evidence to show that it is the other way around?
25. In the above question, let the professor sample ' n ' students and their sum of scores in graduate algorithms is 700. Let the standard deviation of the population be 5. What is the smallest value ' n ' such that the professor rejects the null hypothesis?
26. If you have 50 people in a classroom whose birthdays are assumed to be distributed uniformly at random, then how many pairwise duplicate birthdays do you expect?
27. In the above question, if you have 25 people, then with how much probability you hope to see at least one collision (use Markov)?
28. **[New]** The birthday paradox is that, in a set of n (small integer) randomly chosen people, at least two will share a birthday (dd/mm) with a high probability. Particularly, it refers to the counterintuitive fact that only 23 people are needed for that probability to exceed 50%.
 - a. Now, consider that you have set of people whose age ranges from 10yrs to 20yrs and you are noting their birthdays as dd/mm/yyyy. Now, how many people will you need so that two of them will have the same birthdays with at least 50% probability?
 - b. In the above question, how many people do you need to ensure that the collision happens with at least 75%?
 - c. Now, along with the birthday, every person shouts another random day from the same range of dates. Between the two dates you only consider the date that has no collision with previous considered dates. How many people you will be now to get a collision with atleast 50% probability?

29. Explain the concept of "degrees of freedom" in the context of a one-sample t-test. Why is it $n-1$?
30. A researcher wants to test if a new drug improves patient recovery time. They measure recovery time for 10 patients before the drug and the same 10 patients after the drug. Which type of test should they use and why?
31. What are the limitations of using a t-test for a correlation coefficient?
32. What is the purpose of Maximum Likelihood Estimation (MLE)?
33. What is the main potential drawback of MLE?
34. How does Maximum a posteriori (MAP) estimation differ from MLE?
35. When does the MAP estimate converge to the MLE estimate? Explain why this happens.
36. A medical company touts its new test for a certain genetic disorder. The false negative rate is small: if you have the disorder, the probability that the test returns a positive result is 0.999. The false positive rate is also small: if you do not have the disorder, the probability that the test returns a positive result is only 0.005. Assume that 2% of the population has the disorder. If a person chosen uniformly from the population is tested and the result comes back positive, what is the probability that the person has the disorder?
37. Using Markov's inequality, derive Chebyshev's inequality.
38. Provide an example of a situation where Chebyshev's inequality gives a weak bound.
39. Why is Chebyshev's bound so much better in the coin flip example discussed in class?
40. The number of daily visitors to a website has a mean of 1000. Use Markov's inequality to provide an upper bound on the probability that the number of visitors tomorrow will be at least 3000. Now, you're told the standard deviation is 200. Use Chebyshev's inequality to provide an upper bound on the same probability.
41. For two independent random variables X and Y , prove that $\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y]$.
42. A social media company claims that on average, a user sees no more than 10 ads per day. You monitor 100 users for a day and find that the total number of ads seen across all users is 1500. Can you use Markov's inequality to cast doubt on the company's claim?
43. Why is the minimum over t essential in Chernoff's bound?
44. What are the advantages and disadvantages of using Bloom filters?
45. In terms of false positive and false negative what is the ideal setup to use a Bloom Filters to query a membership problem?
46. Does the False positive rate always decrease with an increasing number of hash functions in a bloom filter?
47. In what way is a Bloom filter a "probabilistic data structure"?

48. Consider a huge dataset of userid, shared between two machines M1 and M2. M1 uses one hash function, h_1 , to construct a bloom filter B1 for the dataset D of size m bits. M2 uses another hash function, k_1 to construct a bloom filter B2 for the dataset D of size m bits.
Let $C = B1 \text{ bitwise-AND } B2$ and $D = B1 \text{ bitwise-OR } B2$.
- Is C also a well defined bloom filter? If yes then what are (or is) its hash functions, else why this is not a bloom filter?
 - Is D also a well defined bloom filter? If yes then what are (or is) its hash functions, else why this is not a bloom filter?
 - If B1 uses two hash functions h_1 and h_2 and B2 uses two different hash functions k_1 and k_2 , then answer a and b.
49. Let U be a set of ' r ' unique elements, out of which we encounter ' s ' elements such that $s \ll r$. If we construct a bloom filter of size ' t ' bits using k hash functions $h_1, h_2, \dots, h_k$ such that its false positive rate is optimal, then answer the following.
- What is the size of t ?
 - What is the value of k ?
 - What is the optimum false positive rate?
50. **[New]** Consider a Bloom filter where exactly one third of the bits are set to 1, and the rest are 0. Suppose we query an element w that has not been inserted into the filter. If the implementation uses k independent, random hash functions, what is the probability that $\text{query}(w)$ yields a false positive.
51. If the stream of the dataset is skewed, that is, some elements appear much more frequently than others, then is FM still a good algorithm to estimate the number of unique elements? Why or why not?
52. Let U be a universe with m unique elements. Recall that a variant of FM hashes $h: U \rightarrow [0, 1]$ such that if s is the smallest hash value that FM algo maintains by the end of the stream, then $E[s] = 1/(m + 1)$. Then is $E[1/s - 1] = m$?
53. Each entry of a JL matrix of dimension $k \times d$ is a sample sample from $N(0, 1/k)$. If we sparsify the matrix such that $4/5$ entries are 0, then from which distribution do we need to sample entries to ensure the desired properties from the sparse JL matrix?
54. What is the expected squared distance between two points generated from the surface of a unit d -dimensional sphere centered at the origin?
55. Explain how random projection can be used to solve linear regression approximately but quickly.
56. For the previous question, explain the merits and demerits of your solution.
57. Let X be matrix of size $2^m \times d$ representing 2^m points in d dimensional space. Let H_m be an $2^m \times 2^m$ be a Hadamard matrix then such that $H_1 = [1]$ and $H_m = \begin{pmatrix} H_{m-1} & H_{m-1} \\ H_{m-1} & -H_{m-1} \end{pmatrix}$.

- a. For a fixed m , define H_m with appropriate scaling such that H_m is an orthonormal matrix.
 - b. Let D be an $2^m \times 2^m$ size diagonal matrix, such that every diagonal value is either $+1$ or -1 with an equal probability. Then what is the effect of DH_m on X (i.e., the 2^m points in $DH_m X$ relates with the points in X in terms of their square of L2 norm)?
58. Define the pseudo inverse of a matrix X of size $n \times m$?
59. Let X be a dataset consisting of n points in d dimensional space and y be a vector of in $\mathbb{R}^n$ representing the targets for every points in the dataset. Let, the data is closely approximated by k -dimensional subspace in $\mathbb{R}^n$. Using a JL matrix M of size $d \times k \log(k)$ design an algorithm to quickly train a model v in $\mathbb{R}^d$.
60. Let x be a random vector on the surface of a unit sphere in $\mathbb{R}^d$.
 - a. Prove that the matrix $E[xx^T] = I$.
 - b. What is the rank of $E[xx^T]$ and xx^T and why?
61. Let $p = \{p_1, \dots, p_n\}$ be a distribution on n points that lie on $\mathbb{R}^d$. Consider X be the data matrix. Write a pseudo code to generate a matrix S that randomly selects m points out of n . Let Y be the sampled points from X using S such that for any vector v in $\mathbb{R}^d$ we get, $E[\|Yv\|_2^2] = \|Xv\|_2^2$
62. When Misra-Gries does not return the count of an element then what can be concluded about the frequency of the element?
63. When a sampling based algorithm for fast matrix multiplication is not useful.
64. Recall that eigenvalue decomposition is used to compute the eigenvalues and eigenvectors of a square matrix. It takes $O(d^3)$ for a $d \times d$ square matrix. Let A be a rectangular matrix of size $n \times m$. Using the eigenvalue subroutine design, an algorithm to compute the singular value decomposition of A in $O(nm \min(m,n))$?
65. Let A and B be two matrices of size $m \times n$ and $n \times p$, respectively. Let $C = AB$.
 - a. Design an algorithm using a random projection that returns D such that $E[D] = C$.
 - b. Propose a variant of the algorithm such that the $\text{Var}(D_{ij})$ is small, where D_{ij} is the entry corresponding to the i^{th} row and j^{th} column of D .
 - c. Design an algorithm using random sampling that returns D such that $E[D] = C$.

- d. Derive the sampling probability such that the following holds and analyze its running time.

$$E[\|D - C\|_F^2] \leq \frac{\|A\|_F^2 \|B\|_F^2}{m}$$

- e. Design a sampling probability that gives a better bound than the above.
66. **[New]** Why is a strong (ϵ, δ) -approximation and how is it useful in training a model?
67. **[New]** Why and how the ϵ -net argument is used in a strong (ϵ, δ) -approximation?
68. **[New]** What is the expected squared distance between two points generated at random inside a d -dimensional cube centered at the origin, having a side length of 2?
69. **[New]** Suppose we have five events E_1, E_2, E_3, E_4, E_5 with probabilities p_1, p_2, p_3, p_4, p_5 . What is the probability that none of these five events occur? In particular, give a tight lower bound for this probability without assuming independence.